Table 3-6. Coverage of Student Outcomes 1-7 by non-ChE technical courses.

Course	**(1)** an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	**(2)** an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	**(3)** (an ability to communicate effectively with a range of audiences.	**(4)** an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	**(5)** an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	**(6)** an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies**.**
MATH 180	4	2	2	1	1	1	1
MATH 181	4	2	2	1	1	1	1
MATH 210	4	2	2	1	1	1	1
MATH 220	4	2	2	1	1	1	1
PHYS 141	4	2	2	3	3	3	1
PHYS 142	4	2	2	3	3	3	1
CHEM 122/123	4	2	1	3	3	4	1
CHEM 124/125	4	2	1	3	3	4	1
CHEM 222	4	2	1	3	3	4	1
CHEM 232	4	2	1	1	1	1	1
CHEM 233	4	2	1	4	4	4	1
CHEM 234	4	2	1	1	1	1	1
CHEM 342	4	2	1	1	1	1	1
CHEM 346	4	2	1	1	1	1	1
CS 109	2	2	4	1	1	1	1
CME 260	4	4	3	3	3	3	1
ECE 210	3	3	3	1	1	4	1

4 = topic is main focus of course, grade largely dependent on demonstrated knowledge;
3 = course grade partially dependent on demonstrated knowledge on this topic;
2 = topic introduced in course, course grade not dependent on demonstrated knowledge;
1 = topic not covered specifically in course.

Abbreviated, descriptive course titles:

MATH 180 Calculus I
MATH 181 Calculus II
MATH 210 Calculus III
MATH 220 Intro. Differential equations
PHYS 141 General Physics I
PHYS 142 General Physics II
CHEM 122/124 Gen Chem I
CHEM 124/125 Gen Chem II
CHEM 222 Analytical Chemistry
CHEM 232 Organic Chemistry I
CHEM 233 Organic Chem. Lab I
CHEM 234 Organic Chemistry II
CHEM 342 Physical Chemistry I
CHEM 346 Physical Chemistry II
CS 109 Prog with MatLab
CME 260 = Properties of Materials ECE 210 = Elect Circuit Analysis ENGR 100 = Engineering Orientation